GW170817 confines the vestigial-second-sound graviton identification to one part in 10^{15} of its natural range

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We constrain our assumed leading-order graviton speed $c_{\rm GW} = c\sqrt{\chi_{\rm vest}}$, where c is the electromagnetic reference speed. The conservative GW170817 bound $|\Delta c/c| \le \tau = 3 \times 10^{-15}$ confines the nonnegative susceptibility to $[(1-\tau)^2, (1+\tau)^2]$, of width 4τ. This interval contains $\chi_{\rm vest} = 1$ and is over fourteen orders of magnitude narrower than our adopted range $[0.1, 10]$. The mode and range are not derived from microscopic dynamics; the constraint applies to this kinematic model, not to emergent gravity generally.

INTRODUCTION

Spinor-gravity constructions use composite vielbeins built from fermions [6, 8]. Volovik proposes a vestigial phase with metric order preceding a nonzero tetrad expectation [1]. His symmetry-breaking analysis distinguishes absorbed Nambu–Goldstone modes from the Higgs-sector gravitational-wave modes, whose counting also uses coordinate constraints [2]. Here we assume that a metric-channel excitation of such a substrate can be identified with a spin-2 second-sound graviton.

Motivation and assumed speed law. Volovik applies Landau two-fluid hydrodynamics to a de Sitter state in [3]. The underlying thermodynamic framework treats the gravitational normal component as stiff matter [4]. Within that de Sitter model, the Landau formula and the stiff-component thermodynamic relations give $s_2 = c$, independently of H. He discusses several possible graviton interpretations of this collective mode and leaves its type open. For the spin-2 substrate identification assumed here, we also assume the susceptibility-dependent speed law:

$$c_{\rm GW}^2(\chi_{\rm vest}) = c^2 \cdot \chi_{\rm vest}, \qquad c_{\rm GW}(\chi_{\rm vest}) = c\sqrt{\chi_{\rm vest}}, \quad (1)$$

where $\chi_{\rm vest}$ is the dimensionless, normalized metric-channel response of the assumed vestigial substrate. It is a free parameter in this calculation. The RPA form in `VestigialSusceptibility.lean` models it using unevaluated bare susceptibility and coupling inputs, so that form alone supplies no numerical value of $\chi_{\rm vest}$. The underlying lattice-fermion framework is the one whose unitarity Vergeles establishes [10]. Equation (1) is an assumed kinematic speed law. We constrain its free susceptibility without deriving a propagating second-sound mode from a microscopic action, its propagator, or its coupling to matter. This missing dynamical bridge is a limitation of the present calculation, not an impossibility result.

The 2017 binary-neutron-star merger GW170817 [11] yielded a tight propagation constraint under the collaboration's conservative distance and emission-time assumptions. Comparing the gravitational-wave arrival time at LIGO/Virgo with the gamma-ray signal at Fermi/INTEGRAL gives $-3 \times 10^{-15} \le \Delta c/c \le +7 \times 10^{-16}$ under those assumptions. We use its conservative symmetric envelope, $|\Delta c/c| \le \tau$ with $\tau = 3 \times 10^{-15}$. This cap inherits the assumed source-emission window; alternative emission histories can weaken the inferred propagation constraint.

Our assumed speed law satisfies the GW170817 bound only on a sub-interval of our adopted range $\chi_{\rm vest} \in [0.1, 10]$ narrower than that range by more than 14 orders of magnitude. The GW170817-compatible window $[(1-\tau)^2, (1+\tau)^2]$ has width 4τ in $\chi_{\rm vest}$ and sits strictly interior to our adopted range $[0.1, 10]$ about the anchor $\chi_{\rm vest} = 1$. Its relative width is $4\tau/9.9 \approx 1.2 \times 10^{-15}$; a probability mass additionally requires a specified prior. The result is a fine-tuning constraint on one specific kinematic identification, not a refutation of emergent gravity nor of the ADW program; we discuss three Phase 6 recovery paths in the Discussion.

Relation to the 2017 constraint sweep. Using GW170817 to eliminate a gravity model is not new. Within months of the detection, four Letters analyzed constraints on specified gravitational actions and cosmological models. Baker *et al.* [12] constrained scalar-tensor and vector-tensor theories and obtained a conditional graviton-mass bound in bimetric gravity. Creminelli and Vernizzi [13] required luminal propagation to remain robust on nearby cosmological backgrounds to avoid tuning, reducing the three independent quartic and quintic beyond-Horndeski functions to one. Sakstein and Jain [14] combined the GW bound with cluster and stellar tests to constrain beyond-Horndeski parameters; their separate combination of GW and M87 black-hole constraints made the covariant quartic galileon cosmologically irrelevant for the order-unity matter couplings considered. Ezquiaga and Zumalacárregui [15] also discussed constraints on GW-speed-modifying couplings in Einstein-aether, some sectors of Hořava gravity, and Generalized Proca theories, and the impact on TeVeS and MOND-like theories.

These four analyses relate GW propagation to specified actions or EFT operators, with assumptions about

backgrounds and matter coupling. They restrict combinations of functions or couplings, mass parameters, and model cosmologies. This Letter supplies no microscopic Lagrangian for the vestigial-second-sound identification. As the Introduction notes, $c_{\rm GW} = c\sqrt{\chi_{\rm vest}}$ is assumed kinematically, so this calculation does not constrain microscopic operator coefficients. What the bound constrains is $\chi_{\rm vest}$: the compatible interval has roughly one part in 10^{15} of the width of our adopted order-unity window. This is an interval-width comparison, not a microscopic prediction of $\chi_{\rm vest}$ or a probability statement independent of a prior. The 2017 sweep constrained *theories*; this constrains an *identification*, which is why it is stated as a ratio of two interval widths rather than as a surviving region of parameter space.

At fixed assumed Eq. (1), this calculation constrains $\chi_{\rm vest}$ rather than microscopic operator coefficients. It does not analyze whether the operator-level mechanisms discussed in those theories alter the effective speed law or its applicability. A microscopic realization would require that separate analysis. The three recovery paths in the Discussion can supply a mechanism fixing $\chi_{\rm vest}$ within the compatible interval, or modify the identification or its leading-order assumptions. A fixed-point mechanism can preserve the speed law; the finite-tolerance constraint alone derives none of these mechanisms.

FORMAL SETUP AND RESULT

Vestigial-susceptibility natural range. Our substrate models the vestigial-phase metric-channel susceptibility by the RPA *resummation* $\chi_{\rm RPA} = \chi_0/(1-\gamma_\star\chi_0)$ in the standard Hertz–Millis convention. Here χ_0 is the bubble integral; $\chi_{\rm RPA}$ is the resummed response and $\gamma_\star$ the effective coupling; physical normalization remains undetermined. `VestigialSusceptibility.lean` formalizes that closed form and its stability properties; it evaluates neither χ_0 nor $\gamma_\star$, so it fixes no numerical $\chi_{\rm vest}$, which is why $\chi_{\rm vest}$ enters Eq. (1) as a free parameter. The natural range $\chi_{\rm vest} \in [0.1, 10]$ is a *project-adopted* naturalness window: the one-loop bubble normalization is $\mathcal{O}(\Lambda_{\rm UV}^2/16\pi^2)$ by dimensional analysis, and we take the $\pm$1-decade window about the resulting unit-normalized central value. We emphasize the provenance: the window is a modeling choice of this project, not a published derivation: the Vergeles lattice analysis [10] establishes unitarity of the underlying Einstein–Cartan–Palatini lattice theory but does not itself compute $\chi_{\rm vest}$ or assign it a range. The window matches the linearized-EFE wave's ADW microscopic-coefficient $\alpha_{\rm ADW} \in [0.1, 10]$ natural range [16], which we treat as the project-internal anchor for the order-of-magnitude prior.

The leading-order $c_{\rm GW}$. Under the assumed law (1), a positive susceptibility and physical reference speed $c > 0$ give $c_{\rm GW} > 0$. For every $c \neq 0$, the relative speed deviation is

$$\Delta c/c(\chi_{\rm vest}) \equiv \frac{c_{\rm GW}(\chi_{\rm vest}) - c}{c} = \sqrt{\chi_{\rm vest}} - 1.$$

On the nonnegative domain this deviation is strictly increasing and vanishes if and only if $\chi_{\rm vest} = 1$. Thus unity is the exactly luminal anchor; finite-tolerance compatibility requires an interval rather than this single value.

Finite-tolerance compatibility. Fix $\tau = 3 \times 10^{-15}$ and write $\texttt{LigoSatisfied}(\delta) \equiv |\delta| \le \tau$ for the conservative symmetric cap.

Theorem 1 (`c_GW_match_iff_chi_close_to_one`)**.** *For* $\chi_{\rm vest} \ge 0$, $\texttt{LigoSatisfied}(\Delta c/c(\chi_{\rm vest})) \iff \chi_{\rm vest} \in [(1-\tau)^2, (1+\tau)^2]$.

Indeed, the cap is equivalent to $1-\tau \le \sqrt{\chi_{\rm vest}} \le 1+\tau$. Since $0 < \tau < 1$, both bounds are nonnegative, so squaring gives the stated interval. Conversely, monotonicity of the square root recovers the cap from these squared bounds.

Every compatible susceptibility therefore satisfies

$$0.1 < (1-\tau)^2 \le \chi_{\rm vest} \le (1+\tau)^2 < 10.$$

The compatible interval contains $\chi_{\rm vest} = 1$ and has nonzero width exactly $(1+\tau)^2 - (1-\tau)^2 = 4\tau = 1.2 \times 10^{-14}$. The adopted range has width 9.9, and the strict comparison is

$$10^{14}\left[(1+\tau)^2 - (1-\tau)^2\right] < 9.9.$$

This is a comparison of interval widths within an adopted domain. It supplies no probability without a prior: under a uniform prior on $[0.1, 10]$, the compatible interval has probability mass $4\tau/9.9$.

Endpoint checks. The endpoints provide quantitative checks of the confinement.

Theorem 2 (`natural_lower_violates_ligo`)**.** *At* $\chi_{\rm vest} = 0.1$, $\Delta c/c = \sqrt{0.1} - 1 \approx -0.6838$, *hence* $|\Delta c/c|/\tau \approx 2.28 \times 10^{14}$ *and* `LigoSatisfied` *is false.*

Theorem 3 (`natural_upper_violates_ligo`)**.** *At* $\chi_{\rm vest} = 10$, $\Delta c/c = \sqrt{10} - 1 \approx +2.1623$, *hence* $|\Delta c/c|/\tau \approx 7.21 \times 10^{14}$ *and* `LigoSatisfied` *is false.*

The exclusions already follow from $\sqrt{1/10} \le \sqrt{1/4} = 1/2$ and $\sqrt{10} \ge \sqrt{4} = 2$. Both endpoints therefore fail simultaneously, while the finite interval in Theorem 1 remains compatible. Figure 1 shows this distinction.

Compatibility of the identification conditions. The proposed identification imposes positivity (P1), compatibility with the cap (P2), and sub-half-deviation (P3'):

$$\begin{aligned}&\texttt{H_VestigialModeIsGraviton}(\chi_{\rm vest})\\ &\quad\equiv \chi_{\rm vest} > 0 \ \wedge\ \texttt{LigoSatisfied}(\Delta c/c(\chi_{\rm vest}))\\ &\qquad\wedge\ |\Delta c/c(\chi_{\rm vest})| < 1/2.\end{aligned}$$

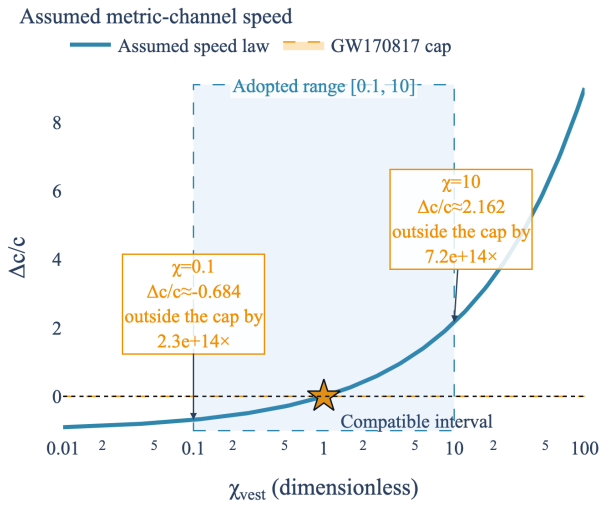


FIG. 1. **Headline result.** The $\Delta c/c = \sqrt{\chi_{\rm vest}} - 1$ deviation across the natural $\chi_{\rm vest} \in [0.1, 10]$ range (pale blue band, dashed blue border) versus the conservative symmetric envelope of the GW170817 [11] multi-messenger constraint, $|\Delta c/c| \le 3 \times 10^{-15}$ (amber, drawn at $\Delta c/c = 0$ because the cap's own width is far below one pixel at this scale). Over the natural range $\Delta c/c$ sweeps $[-0.68, +2.16]$, exceeding the cap by $\sim 10^{14}\times$ *at the endpoints* (orange callouts) and vanishing at $\chi_{\rm vest} = 1$. The compatible $\chi_{\rm vest}$ window $[(1-\tau)^2, (1+\tau)^2]$ has width $4\tau = 1.2 \times 10^{-14}$, compared with the adopted range's width 9.9. Its finite extent is unresolved at this scale; the gold star marks its anchor $\chi_{\rm vest} = 1$, not a singleton compatible set. The window sits *inside* the band: the datum confines $\chi_{\rm vest}$, it does not exclude the range.

These three clauses are not independent. At the fixed tolerance, P2 implies P3' because $\tau < 1/2$. On the nonnegative domain, Theorem 1 also places every P2-compatible susceptibility in a strictly positive interval, so P1 holds there. Consequently, the complete compatible set for these conditions is precisely that same finite interval, including the anchor $\chi_{\rm vest} = 1$.

Four counterexamples distinguish compatibility from positivity alone. The endpoints 0.1 and 10 refute the conjunction through failure of P2. At zero, P1 fails; this point lies outside the adopted range. At the interior point $\chi_{\rm vest} = 1/4$, $\Delta c/c = -1/2$: P1 holds, but P2 and the strict P3' inequality both fail. The quarter-point example therefore shows that P1 alone does not imply P3'. These counterexamples do not establish independence of the three clauses or isolate a uniquely failing clause. The corresponding definitions and algebraic results are recorded in `GravitationalWaves.lean`, Sections 1–4.

Endpoint constraints and the open dynamical bridge. The endpoint exclusion also applies directly to the relative deviation of the speed itself:

$$\left|\frac{c_{\rm GW}(\chi_{\rm vest}) - c}{c}\right| > \tau, \qquad \chi_{\rm vest} \in \{0.1, 10\}, \quad c \neq 0.$$

This is the statement of `natural_endpoints_violate_ligo_for_every_c`; it includes the physical domain $c > 0$ and follows from the relative-deviation identity above. Changing c therefore cannot make either endpoint compatible under the assumed law. This two-point exclusion leaves the compatible interior interval intact. Neither it nor the c-free endpoint statement quantifies over microscopic actions, propagators or matter couplings, so neither establishes the absence or impossibility of a dynamical derivation. A physical application still requires the response normalization and domain, the physical state, a propagating tensor mode relative to the same matter/light reference, and controlled transport and frequency matching with units and error bounds.

DISCUSSION

What this constrains. The conservative symmetric cap restricts the assumed leading-order identification to $\chi_{\rm vest} \in [(1-\tau)^2, (1+\tau)^2]$. Expanding the endpoints gives $1 - 2\tau + \tau^2$ and $1 + 2\tau + \tau^2$: the susceptibility offsets are approximately $\pm 6 \times 10^{-15}$, whereas the speed-deviation tolerance is $\pm 3 \times 10^{-15}$. Exact luminality occurs at $\chi_{\rm vest} = 1$, but the finite measurement tolerance permits a nonzero-width interval. Its small relative width within our adopted range does not alone establish a prior probability, rule out generic UV theories, or require an exact fixed point at unity.

What this does not rule out. Three Phase 6 paths remain open:

1. **Symmetry-driven $\chi_{\rm vest} \to 1$ fixed point.** If a deep-IR symmetry (e.g. Lorentz invariance of the emergent metric sector) makes $\chi_{\rm vest} = 1$ an exact fixed point of the RG flow, the fine-tuning is structural rather than tuned. An exact fixed point at $\chi_{\rm vest} = 1$ would lie inside the compatible interval. The finite-tolerance constraint alone does not select this mechanism or establish that a microscopic model realizes it. Phase 6 should examine whether such a fixed point is realized in the ADW broken phase.

2. **Distinct metric-channel identification.** We propose as an alternative that the "vestigial mode" is a separate UV-input metric channel rather than the second-sound DOF. Its dynamics could in principle pin its susceptibility to unity; this remains a model-building possibility. The missing dynamical bridge in this Letter does not establish that this

alternative is more likely. Phase 6e (full nonlinear emergent action) is the natural venue for the disambiguation.

3. **Frequency-dependent $\chi_{\rm vest}$.** Equation (1) is an assumed frequency-independent kinematic law. A frequency-dependent susceptibility or a compensating correction in the LIGO band is a possible phenomenological extension. We consider the algebraic ansatz $\delta_{\rm disp} = \gamma\omega$, without asserting that it is the leading term of a matched microscopic derivative expansion. The formal probe is numerical $\omega = 100$; interpreting it as a 100 Hz band and setting $\gamma = \Gamma_H/c_{\rm GW}^2$ are modeling assumptions requiring physical normalization and microscopic matching. `vestigial_dispersion_below_ligo_at_inspiral_peak` assumes $|\gamma| \le 10^{-30}$ and proves that $|\delta_{\rm disp}|$ at that numerical probe is below the GW170817 tolerance. The adopted 10^{-30} is a placeholder chosen to keep the correction below the cap; it is neither derived nor measured. In the further additive model $\delta_{\rm tot} = \delta_0 + \delta_{\rm disp}$, with $\delta_0 = \sqrt{\chi_{\rm vest}} - 1$, that assumed coefficient bound gives $|\delta_{\rm disp}| \le B = 10^{-28}$ at $\omega = 100$. The reverse triangle inequality then excludes restoration of $|\delta_{\rm tot}| \le \tau$ whenever $|\delta_0| > \tau + B$. Offsets just outside $\pm\tau$ can nevertheless become compatible through an oppositely signed correction within this bound. Larger corrections are not excluded by the placeholder theorem. For a positive numerical probe ω, `dispersion_within_ligo_iff` gives $|\gamma\omega| \le \tau \iff |\gamma| \le \tau/\omega$ at the fixed GW170817 tolerance τ, bounding the correction alone. `dispersion_correction_abs_bound`, `dispersion_correction_linear_in_gamma` and `dispersion_correction_zero_at_no_dissipation` describe the product's magnitude, additivity in γ and zero-coefficient limit. Deriving Γ_H and the normalization of this ansatz from microscopic dynamics remains open.

Comparison to the linearized-EFE wave. The companion linearized-EFE wave [16] ships an emergent-gravity identification *compatible* with observation at the natural-range level: the Sakharov–Adler form $G_{\rm N,emerg} = \alpha_{\rm ADW} \cdot 12\pi/(N_f\Lambda_{\rm UV}^2)$ at $\Lambda_{\rm UV} = M_{\rm Pl}$ produces $\alpha^\star_{\rm ADW} \in [0.40, 1.27]$ for SM N_f, sitting inside the natural range $[0.1, 10]$. The asymmetry between the two waves is a feature of the program: the linearized-EFE $G_{\rm N,emerg}$ depends on the Sakharov–Adler kernel computed at the linearized level, where the relevant scale is $M_{\rm Pl}$; $c_{\rm GW}$ depends on the free metric-channel susceptibility $\chi_{\rm vest}$. Our adopted interval contains unity but derives no value of this susceptibility. The conservative GW170817 cap restricts it to the much narrower compatible interval; the width comparison alone supplies no microscopic selection mechanism.

Reproducibility. The result is reproducible in two layers. The Lean layer is `GravitationalWaves.lean` (32 theorems, zero `sorry`, zero new axioms, zero `maxHeartbeats` overrides) at the project's pinned Mathlib4 commit; the Python layer is `src/gravitational_waves/` with 89 `pytest` cases cross-checking every numeric quoted in the abstract and figure caption. The headline figure (`src/core/visualizations.py:fig_c_GW_vs_ligo_constraint`) and its underlying numerics are committed-pinned.

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